

Regulatory Resilience in Technology Transfer Systems: Relational Contracting and State Capacity in Vietnam

Anonymous manuscript for The Journal of Technology Transfer

Abstract

Technology transfer succeeds when a continuing inter-organisational relationship reaches operating capability while remaining connected to multiple public mandates. This article develops a dual-function account of the technology-transfer contract: it is both a private performance arrangement and a compliance protocol that organises information and regulatory engagement without altering public powers. Vietnam's 2025–2026 reforms provide a theory-informing case. A systematic qualitative analysis of thirty-four Vietnamese policy, professional and legal publications identifies recurring transaction problems; two source-traced vignettes expose their temporal and organisational structure. Doctrinal analysis then tests those problems against technology-transfer, intellectual-property, competition, data, artificial-intelligence and sectoral law. The analysis identifies three connected mechanisms: verifiable delivery and adaptation, cross-regime compliance coordination, and complementary administrative and industrial capability. The statutory framework provides extensive but distributed protection. It does not, however, supply a technology-specific default connecting acceptance, corrective work, payment and warranty, or a routine interface through which firms can obtain coordinated, reviewable regulatory answers. Relational contracting and state capacity are therefore complements: their interaction produces regulatory resilience, which in turn supports transfer performance. A time-limited coordination pilot and two propositions make that causal argument testable. Supplementary text-model diagnostics are retained only as an audit of theme sensitivity, not as statistical evidence from the small corpus. The contribution is a mechanism-based account of regulatory resilience that can be examined beyond Vietnam.

Keywords: technology transfer; regulatory resilience; relational contracts; state capacity; administrative coordination; Vietnam

JEL codes: O31; O32; O38; K12; K21

1 Introduction

Technology transfer is often described as a movement of knowledge or rights from one organisation to another. In practice, the movement is rarely complete at signature. Operating capability may depend on training, trial production, corrective work, access to tacit know-how and repeated decisions about adaptation. The same transaction may also engage intellectual-property, competition, data, artificial-intelligence, investment and sectoral rules. A transfer system is therefore resilient only when it can sustain private performance and connect the transaction to applicable public mandates as technologies and circumstances change.

This problem is especially visible in economies that seek advanced technology while rebuilding their legal and administrative architecture. It is tempting to diagnose every overlap as fragmentation or every missing technology-specific rule as a regulatory void. That diagnosis is too quick. Specialisation across statutes can preserve expertise and accountability, and general private law can supply remedies where a dedicated transfer statute is silent. The analytically useful question is narrower: which connections are needed for a continuing transfer relationship to reach demonstrable operating capability, and which of those connections are supplied by law, contract or administrative practice?

Vietnam provides a theory-informing case rather than a representative case of all emerging economies. It combines a dedicated Law on Technology Transfer with general contract law, intellectual-property and

competition regimes, and recently enacted rules on data, artificial intelligence, digital industries and high technology. Between 2025 and 2026 the legislature expanded the objects of transfer, revised intellectual-property rules, enacted a national artificial-intelligence statute, adopted semiconductor provisions and replaced the principal technology-transfer decree. The reforms make it possible to observe how new sectoral regimes interact with an older contract architecture.

The article begins with published accounts rather than presumed statutory omissions. A systematic qualitative document analysis organises a bounded corpus of thirty-four Vietnamese publications by source type, transaction problem and institutional function. The documents identify questions; they do not establish population frequencies, causal effects or the prevalence of defective contracts. Doctrinal analysis then asks where the operative answer to each question is located, under what conditions it applies and whether a tailored default or connection remains unspecified. This sequence preserves the distinct evidential roles of documentary interpretation and legal analysis.

Three research questions guide the study. First, which performance, compliance and capability questions emerge from the principal themes in the selected discourse? Second, which Vietnamese rules answer those questions, and where does the review identify no tailored default or routine regulatory connection? Third, what does the resulting pattern imply for the resilience of technology-transfer systems and for subsequent empirical evaluation?

The central contribution is a dual-function account of the technology-transfer contract. The contract organises delivery, acceptance, training, adaptation and payment between private parties. It can also organise the records, information duties, role allocation and cooperation needed to engage public authorities. The second function does not make regulators parties to the contract or permit private agreement to alter mandatory duties. It treats the transaction record as connective infrastructure between distinct legal functions.

The analysis identifies three mechanisms. The first is verifiable performance over time: a transfer must connect demonstrable delivery to correction, payment and warranty. The second is cross-regime compliance coordination: recognising an algorithm or dataset as transferable does not determine ownership, lawful access, security duties or downstream AI responsibilities. The third is complementary capability: access to a patent or model cannot substitute for skills, facilities, finance, valuation expertise or administrative delivery. These mechanisms explain why more detailed documentation can be necessary without being sufficient.

The article proceeds as follows. Section 2 develops the theoretical framework and contribution. Section 3 explains the qualitative document analysis, supplementary audit and doctrinal scope. Section 4 presents the three mechanisms through an evidence matrix and source-traced vignettes. Section 5 states the dual-function framework and a coordination pilot. Section 6 discusses transferability, rival explanations and testable implications. Section 7 concludes.

2 Theory and contribution

Relational contract theory directs attention from isolated promises to continuing cooperation. Macaulay (1963) showed that business exchange relies on practices and relationships that exceed formal terms. Macneil (1980) treated contractual obligations as embedded in continuing social relations. Williamson (1979) offered a distinct transaction-cost explanation in which uncertainty and transaction-specific investment create governance problems after parties commit resources. These approaches should not be collapsed, but together they identify adaptation, information exchange and safeguards against opportunism as central to transactions whose performance cannot be specified completely in advance.

Innovation contracting sharpens that insight. Gilson, Sabel and Scott (2009) explain how formal arrangements can support iterative collaboration rather than simply define a finished product. A detailed document and a

well-governed relationship are not alternatives. The relevant question is which provisions make experimentation, verification and correction workable. For technology transfer, that includes specifications of operating capability, access to tacit knowledge, training deliverables, testing responsibilities, change procedures and independent expert determination.

Evidence accumulated since that contribution reinforces a complementary rather than substitutive view. Cao and Lumineau's (2015) meta-analysis of 149 studies finds that contractual governance is positively associated with trust and relational norms and that the mechanisms jointly improve relationship performance. Recent work on academic engagement likewise treats knowledge exchange as organisationally broader than licensing and patenting (Perkmann et al. 2021). The implication for technology transfer is not that every contingency should be specified *ex ante*, but that formal records and relational routines should be designed to support learning and adaptation together.

The innovation literature also prevents an exclusively contractual account. Teece (1986) links returns from innovation to appropriability and complementary assets. Grindley and Teece (1997) show the importance of negotiated licensing and cross-licensing in semiconductors and electronics. Bradley, Hayter and Link (2013) criticise linear accounts of university technology transfer that neglect informal mechanisms and organisational conditions. Bozeman (2000) and Bozeman, Rimes and Youtie (2015) similarly treat transfer effectiveness as contingent on the actors, objects, mechanisms and public values involved. Clear rights may therefore be necessary without being sufficient where skills, finance, facilities or organisational incentives remain binding constraints.

Administrative capacity connects these private relationships to public organisation. Lodge and Wegrich (2014) distinguish analytical, coordination, regulatory and delivery capacities. Applied to technology transfer, those capacities concern identifying a transaction's legal implications, connecting competent bodies, exercising lawful oversight and supporting implementation. The presence of several statutes or authorities does not prove dysfunctional fragmentation. Regulatory distribution becomes problematic when relevant decisions, information or responsibilities cannot be connected in a manner that permits lawful performance.

This systems perspective is consistent with recent technology-transfer research. A forty-five-year review of *The Journal of Technology Transfer* identifies capacity, sustainable transfer and changing non-market environments as important directions (Abid, Cunningham and Perea-Vicente 2025). A complementary review of 1,944 university-technology-transfer studies shows that processes, enablers and capacity now occupy a central position in the literature (Cunningham, Menter and Starke 2025). Together, these studies explain why a contract-only or university-only account is incomplete: transfer outcomes depend on relationships among firms, intermediaries, public bodies and complementary resources at micro, meso and macro levels.

The article joins these literatures by treating regulatory resilience as a property of connections rather than rule density. A resilient transfer system must preserve the expertise and accountability of specialised regimes while making their interfaces usable for the transaction. Private performance and public compliance remain legally distinct, yet they may depend on shared records, timely role identification and coordinated responses. The proposed dual-function contract is one possible connective device; an inter-agency case protocol is another.

Figure 1 states the causal architecture. Relational contracting reduces transaction-level uncertainty by making delivery, adaptation and information exchange workable. State capacity supplies analytical, coordination, regulatory and delivery functions. Their interaction creates regulatory resilience: usable connections across private performance and public mandates. Regulatory resilience should improve the probability that a transfer reaches operating capability, subject to technological complexity, absorptive capacity and institutional resources. A contract cannot bind a regulator, waive statutory powers or privately negotiate away public duties.

The legal contribution is a scoped map of Vietnamese technology-transfer protection after the 2025–2026 reforms. Industrial-property and competition rules overlap more than a registration-plus-dominance account

implies; digital transfers require coordination among transfer, data and AI statutes rather than a single ownership rule; and technology-specific defaults for acceptance and improvements to unregistered know-how remain unspecified in the dedicated contract chapter. The theoretical contribution is the proposed connection between relational performance and administrative coordination.

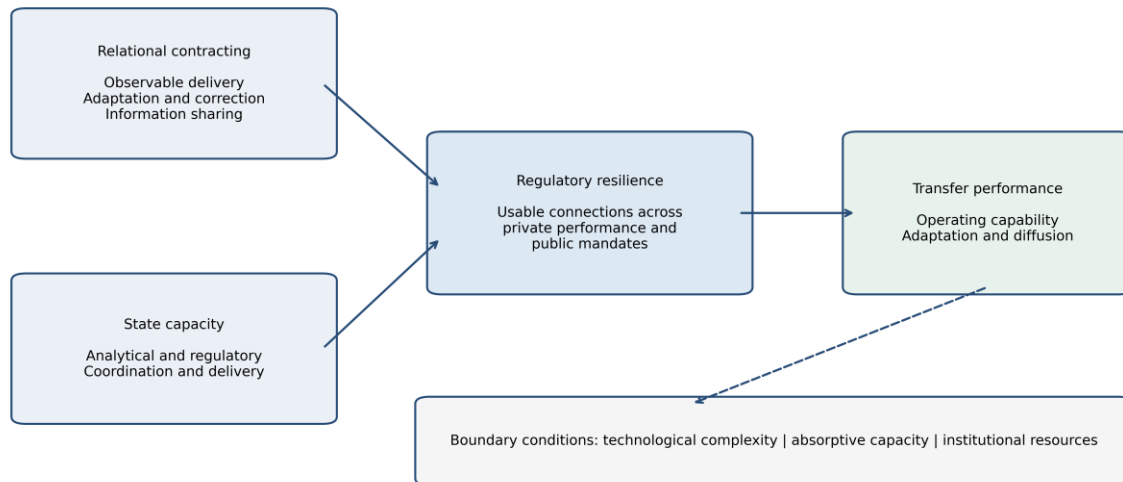


Fig. 1 Causal framework for regulatory resilience and technology-transfer performance

3 Research design

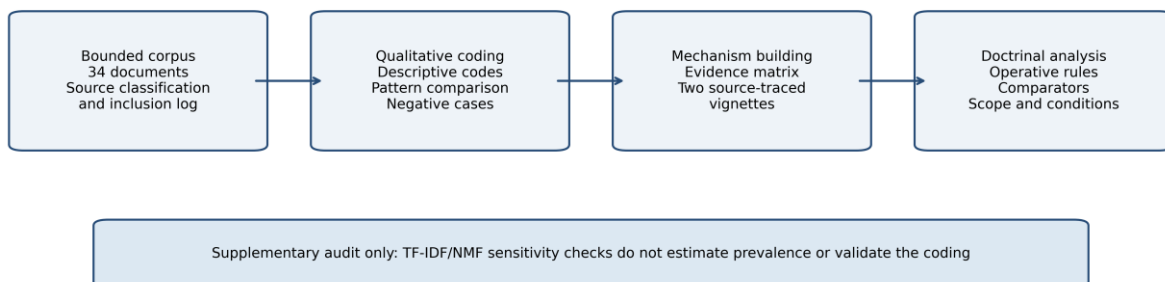


Fig. 2 Research design and evidence boundary

3.1 Evidence strategy and limits

The study uses systematic qualitative document analysis to connect observed transaction problems to doctrinal questions. The design separates four operations: corpus construction and source classification; descriptive coding of the problem each passage reports; pattern coding into candidate mechanisms; and legal assessment against operative instruments. This separation matters because published claims are evidence of discourse and

reported experience, whereas legal conclusions must rest on the text, scope and commencement of applicable rules.

The coding frame was both deductive and inductive. Initial sensitising categories came from relational contracting and state-capacity theory: performance, adaptation, information, regulatory coordination and complementary capability. Close reading then added object-specific codes for acceptance, tacit know-how, restrictive licensing, data access, AI roles, valuation, semiconductor capability and agricultural diffusion. Pattern comparison consolidated those codes into three mechanisms only when the same functional problem appeared across more than one source type or legal interface. Table 2 preserves representative source-to-inference links and a negative or limiting case for each mechanism.

The corpus contains published discussion rather than contracts, judgments or administrative case files. It cannot estimate prevalence, establish causal effects or verify the outcome of a reported dispute. The two vignettes in Section 4 are therefore source-traced illustrations, not adjudicated case studies. No claim of theoretical saturation or inter-coder reliability is made. The anonymized inventory, evidence matrix and supplementary sensitivity outputs permit an audit of selection and interpretation.

Generative AI tools assisted language editing, document restructuring and code review. The authors independently verified the legal sources, qualitative coding, supplementary outputs, analysis and conclusions and retain full responsibility for the final manuscript. No AI system is listed as an author.

3.2 Corpus selection and qualitative procedure

The bounded corpus comprises thirty-four Vietnamese-language documents on technology transfer, science, technology and innovation. Twenty-seven are government, policy or news publications and seven are legal or professional analyses. Documents were included when their principal subject was technology transfer or when they described a concrete transfer setting in AI, semiconductors, agriculture, energy or university research. Duplicative items were retained only when they represented a distinct institutional voice or application setting. The inventory records the filename and operational genre classification; full texts remain in the research archive because most are third-party copyrighted materials.

Table 1 reports the corpus taxonomy rather than topic-model clusters. First-cycle coding recorded the transfer object, actor, reported problem, institutional level and legal interface. Second-cycle comparison asked whether the passage concerned observable performance over time, a connection among regulatory regimes, or a complementary capability needed to use the technology. A passage could support more than one code, but a mechanism was retained only where the distinction changed the relevant legal or policy response.

Validity rests on source tracing, triangulation and explicit limits rather than statistical fit. Contract-performance claims were checked against the dedicated statute and general private-law routes; digital claims were checked across transfer, intellectual-property, data and AI rules; capability claims were checked against sectoral and administrative instruments. Contrary evidence was retained: the existence of substantial general remedies limits any claim of a legal vacuum, while genre sensitivity shows that contract-law concerns are concentrated in professional sources rather than the broader policy discourse.

Text modelling is confined to the Supplementary Information as a sensitivity audit. On this small mixed-genre corpus, NMF, K-means and the K scan cannot support population inference or validate the qualitative coding. Their limited use is to show that the contract-and-dispute vocabulary survives removal of one legal article but disappears from a news-only model. The result supports a genre warning, not a quantitative finding. The former MDS plot has therefore been removed from the manuscript.

Source type	N	Principal coverage	Analytical role
Government, policy and news	27	Transfer programmes; AI; semiconductors; agriculture; energy; local implementation	Maps implementation settings and public problem framing
Legal and professional analysis	7	Contract terms; disputes; IP; competition; services; state management	Identifies legal questions and reported practitioner concerns
Total	34	Full Vietnamese-language documents	Bounded qualitative corpus; not a population sample

Table 1 Corpus taxonomy and analytical role

Mechanism	Representative source trace	Inference	Limiting or rival case
Verifiable delivery	Professional account of obsolescence/payment and incomplete know-how; contract-law analyses	Performance unfolds through trial, acceptance, correction and payment	No case files; frequency and outcome unknown
Cross-regime coordination	Digital-policy documents; transfer, IP, data and AI instruments	Transferability, rights, access and downstream duties are distinct functions	Multiple rules do not by themselves prove fragmentation
Complementary capability	Semiconductor, agriculture and provincial implementation documents	Licences require skills, facilities, appraisal and diffusion channels	Capability constraints may explain outcomes independently of legal design

Table 2 Evidence matrix from sources to bounded inference

3.3 Doctrinal material and comparative scope

The legal review uses 1 October 2026 as its cut-off and distinguishes enactment from commencement. The technology-transfer baseline is Law No. 07/2017/QH14 as amended, read with Consolidated Text No. 138/2026/VBHN-LQ-VPQH and Decree No. 101/2026/NĐ-CP. The review also uses the Intellectual Property Law as amended through Law No. 131/2025/QH15; Competition Law No. 23/2018/QH14; Civil Code No. 91/2015/QH13; Commercial Law No. 36/2005/QH11; and Law on Science, Technology and Innovation No. 93/2025/QH15.

Digital and capability analysis includes Data Law No. 60/2024/QH15, Artificial Intelligence Law No. 134/2025/QH15, the semiconductor chapter of Digital Technology Industry Law No. 71/2025/QH15 and High Technology Law No. 133/2025/QH15. Valuation is read with Prices Law No. 16/2023/QH15, its 2025 amendment and Circular No. 37/2024/TT-BTC. Law No. 20/2026/QH16 was adopted on 24 August 2026. Its Article 4 amendments to the Law on Technology Transfer, concerning responsibility for market-development and technology-import programmes, entered into force on 1 October 2026 under Article 5(2); the balance of Law No. 20/2026/QH16 takes effect on 1 March 2027. The manuscript therefore treats Article 4 as operative and the remaining amendments as prospective.

The international comparison is functional and selective. Relational contracting informs performance questions; US licensing guidance and the EU technology-transfer exemption framework illuminate competitive restraints; the EU Data Act and AI Act distinguish access from downstream compliance; and TRIPS and the US and EU chips instruments distinguish compulsory access from capability building. The sources are not coded into the Vietnamese corpus, do not create a matched cross-country evaluation and do not justify transplantation without regard to institutional powers and resources.

A negative finding is limited to the instruments and function examined. Absence of a technology-specific default does not imply absence of all legal protection. Equivalent duties may arise under general law or

implementing instruments. Conversely, recognising an object as transferable does not establish ownership, lawful access or permission to disclose it.

4 Findings

4.1 Mechanism one: verifiable delivery and adaptation

The largest concentration of documents concerns general transfer activity, innovation, university commercialisation and transfer contracts. Their interpretive connection is continuing performance between identifiable parties. A practitioner account reports disputes across standalone transfer agreements, franchises, assignments of industrial-property rights and equipment sales accompanied by technology transfer. It also states that the parties' rights and obligations depend largely on agreement because the dedicated statute provides only basic allocations. These accounts justify asking what makes delivery and adaptation verifiable; they do not prove that statutory design caused the reported disputes.

Source-traced vignette 1 makes the relational problem concrete. A Vietnamese professional account describes two recurring dispute patterns rather than naming adjudicated cases. In the first, a transferee resists payment after market change by asserting that the technology has become obsolete. In the second, training and trial operation appear complete, but later production fails specified criteria and raises the possibility that critical tacit know-how was withheld. Because the source reports practitioner experience without case files, the article does not infer frequency or outcome. The analytical value lies in the temporal sequence: specification, trial, acceptance, payment, correction and continuing access must be connected if either party is to verify performance.

Article 23 of the Law on Technology Transfer specifies contractual subjects including the technology, transfer mode, rights and obligations, price, schedule, warranty, penalties and liability. Article 27 and Article 18 of Decree No. 101/2026/NĐ-CP supply detailed payment options and percentage bases. These rules are substantial. They nevertheless do not establish a technology-specific acceptance default linking performance tests, corrective work, payment milestones and warranty commencement. This is narrower than a claim that Vietnamese law has no concept of acceptance or no general remedies.

A resilient performance arrangement could specify demonstrable operating capability, training outputs, test conditions, access to necessary tacit knowledge, responsibilities for data and inputs, correction periods, change-control procedures and independent expert determination. It should connect milestone payments and warranty commencement to observable events while protecting confidential know-how. Whether these provisions should take the form of a statutory default, professional model terms or sector-specific guidance remains a policy choice requiring transaction-level evidence.

Restrictive terms illustrate why distributed protection cannot be reduced to a legal vacuum. Intellectual Property Law Article 144(2) prohibits specified unreasonable restrictions in industrial-property licences, and Article 144(3) makes prohibited provisions void. Qualifying trade secrets can be protected without registration. Competition Law Articles 11–14 address restrictive agreements independently of dominance, while Articles 24 and 27 govern abuse of dominance. A non-dominant transferor licensing unpatented technology is therefore not automatically beyond regulation.

The residual question is whether a restrictive provision falling outside an applicable IP prohibition and below a competition-law threshold receives sufficiently predictable assessment under general contract law. The EU technology-transfer exemption and US licensing guidelines show how a system can distinguish restraints that support transfer-specific investment from those that unnecessarily foreclose competition. They do not support making every grant-back, sourcing or territorial restriction void, nor do they justify importing foreign market-share thresholds without institutional analysis.

The theoretical implication is complementary formalisation. Relational governance does not counsel less documentation. It identifies the functions that documents must perform: make delivery observable, preserve capacity for adaptation, allocate information responsibilities and provide a credible route for resolving technical disagreement.

4.2 Mechanism two: cross-regime compliance coordination

Digital-technology documents frame transfer in the vocabulary of security, sovereignty, data protection and control. The legal task is to separate four functions: identifying an object of transfer, determining rights in it, regulating access or movement, and allocating duties when an AI system is developed, supplied or deployed. Combining those functions into a single question of ownership obscures rather than resolves the transaction.

The amended Law on Technology Transfer expressly includes models, algorithms, software, information and data. That recognition is significant but limited. It does not make every dataset proprietary, authorise disclosure, override personal-data or cybersecurity rules, or determine the responsibilities of downstream AI actors. Intellectual-property protection varies with the object: a computer program, an original compilation, a qualifying trade secret and raw data do not attract identical rights.

The Artificial Intelligence Law No. 134/2025/QH15 differentiates developers, providers and deployers, classifies systems by risk and imposes transparency and incident-related obligations. The Data Law separately governs data, including cross-border transfer and processing. Decree No. 101/2026/ND-CP issues encouraged, restricted and prohibited technology lists. Whether a model, dataset or associated know-how falls within a list or another sectoral restriction requires transaction-specific classification. The evidence reviewed here does not establish a categorical screening vacuum.

The EU Data Act offers a functional comparator because it creates specified access arrangements for connected-product and related-service data while preserving personal-data and trade-secret protections. The EU AI Act addresses a different function through provider documentation, downstream information, copyright-compliance and training-summary duties for defined general-purpose models. Neither instrument creates a general ownership right in data or a compulsory licence to all training material.

The implication is coordinated guidance rather than statutory consolidation for its own sake. Parties need to identify their regulatory roles, lawful access, permitted uses, onward disclosure, security responsibilities, records, incident cooperation and change notification. A contract can allocate operational responsibility and require cooperation, but mandatory duties remain mandatory and competent authorities retain their powers.

This mechanism connects private and public performance. The information needed to demonstrate contractual delivery may overlap with information needed to classify a system or respond to an incident. A protected transaction record can reduce repeated requests and inconsistent descriptions, provided access, purpose, retention and review rights are legally controlled.

Source-traced vignette 2 concerns provincial project implementation. A current administrative account reports that some projects describe technology mainly as imported machinery while giving less attention to process knowledge, training and the recipient's ability to operate it. It also reports that technology appraisal can become procedural and that smaller firms struggle to navigate land, tax and support conditions. The source does not identify a single failed project, so it cannot establish administrative causation. It does reveal how the same transaction can fail at two interfaces: information needed for appraisal is not aligned with information needed for operating capability, and support measures are not readily usable by the intended recipient.

4.3 Mechanism three: complementary capability and diffusion

The semiconductor and agricultural themes show why access rights alone cannot explain transfer outcomes. Semiconductor documents emphasise engineers, design and manufacturing capability, national innovation

institutions and Vietnam's position in global value chains. The agricultural documents emphasise extension institutions, farmer adoption, production conditions and organised data. Both themes concern the ability to use technology, but their institutional routes differ.

Vietnam's semiconductor framework extends beyond compulsory licensing. Digital Technology Industry Law No. 71/2025/QH15 contains a dedicated semiconductor chapter, while the high-technology framework and strategic-technology lists address incentives and prioritisation. Intellectual Property Law Article 146(1)(b) nevertheless matters because it limits compulsory licensing of semiconductor patents to public non-commercial use or the remedy of anti-competitive conduct, reflecting TRIPS Article 31(c). A patent licence cannot by itself supply tacit manufacturing knowledge, skilled personnel or facilities.

The US CHIPS and Science Act and EU Chips Act illustrate capability-oriented interventions, including funding, design capacity, pilot lines and competence centres. They are functional comparators, not transplantation templates or general mandates to open patent rights. The inference for Vietnam is to connect sectoral incentives with voluntary licences, training, shared facilities and transparent access terms while keeping treaty-constrained compulsory licensing distinct.

Agricultural diffusion operates through ministries, associations, extension centres and programmes rather than principally through bilateral licensing. Article 52(5) of the Law on Technology Transfer supplies an administrative channel for recognising technical advances encouraged for transfer. The relevant resilience questions concern extension capacity, local adaptation, complementary inputs and user capability. A contract-centred model alone cannot capture them.

Valuation cuts across all three mechanisms. Article 27 distinguishes negotiated price from situations requiring audit or appraisal; Decree No. 101/2026/NĐ-CP and Circular No. 37/2024/TT-BTC supply further standards. The policy priority is usable evidence, transparent assumptions and professional expertise rather than a single compulsory price formula. Cost, market and income approaches each carry limitations where comparables are scarce and future benefits uncertain (Kamiyama, Sheehan and Martinez 2006).

5 A dual-function framework for regulatory resilience

The findings support a dual-function account of the transfer contract. Function 1 is private performance: the contract identifies the technology, links delivery to acceptance, organises training and corrective work, allocates adaptation and improvement rights, and connects observable milestones to payment and warranty. Function 2 is compliance coordination: the contract records the parties' regulatory roles, lawful bases for data and know-how access, information duties, security responsibilities and procedures for regulatory engagement.

These functions are connected but not merged. A contract cannot confer ownership that the law does not recognise, authorise unlawful disclosure, bind an authority or transfer a public power. Nor does administrative registration constitute technical acceptance or commercial viability. The framework instead asks whether information produced for performance can be reused lawfully to make regulatory interfaces intelligible and whether regulatory decisions arrive early enough to permit adaptation.

Table 3 maps ten recurrent questions to the principal legal route. Its categories avoid two opposite errors: treating every referral as a gap and treating the existence of any rule as proof of an effective connection. Transfer-framework rules operate within the dedicated statute; distributed or conditional protection depends on another regime or eligibility condition; an open default-design question identifies the absence of a technology-specific default for the precise function reviewed without excluding general law. Table 4 then compares four functions across selected international instruments. The comparison is diagnostic rather than a claim of legal equivalence: it identifies the design question that each comparator makes visible for Vietnam.

For state management, the proposal is a time-limited coordination pilot rather than a new general approval. A single contact would triage a case and route defined questions to relevant competent bodies. Each body would retain its statutory authority. A protected case record would identify contacts, legal bases, status, reasons and review routes. Scheduled review would examine unresolved referrals alongside acceptance, training and operating capability.

The pilot must be tested for costs as well as benefits. It could reduce repeated information requests and late classification changes, or create another bottleneck. Confidential know-how and personal data must not become generally accessible. Smaller firms must be able to use the process, referrals must remain contestable, and service targets must not displace statutory time limits or substantive tests. Figure 4 states these safeguards as part of the design rather than benefits assumed to follow from a portal.

Question	Principal legal route	Classification
Contract contents and transfer modes	Law on Technology Transfer arts 6, 22–23	Transfer-framework rule
Price and payment	Law art 27; Decree 101/2026 art 18	Transfer-framework rule
Restrictive licence terms	IP Law art 144; Competition Law arts 11–14, 24–27	Conditional protection
Delivery, acceptance and warranty linkage	Contract terms plus applicable general law	Open tailored-default question
Digital transfers and downstream AI duties	Decree 101; Data Law; AI Law	Distributed, transaction-specific
Semiconductor capability	Digital Technology Industry Law ch III; High Technology Law	Distributed regulation
Semiconductor compulsory licensing	IP Law art 146(1)(b); TRIPS art 31(c)	Treaty-constrained protection
State-funded research results	Science, Technology and Innovation Law arts 24–28	Distributed regulation
Agricultural diffusion	Law on Technology Transfer art 52(5)	Administrative transfer channel
Private-transfer valuation	Agreement; Decree 101; valuation standards	Framework plus capability question

Table 3 Selected regulatory interfaces around a technology-transfer relationship

Function	Selected comparator	Vietnamese position	Design implication
Adaptive performance	US/EU contract law and licensing guidance leave delivery architecture mainly to agreement	Dedicated contract chapter lists contents, price, warranty and liability; no technology-specific acceptance sequence	Tie tests, corrective work, payment and warranty to observable milestones
Restrictive licensing	EU TTBER/guidelines and US licensing guidance distinguish efficiency-supporting restraints from foreclosure	IP Law art 144 and Competition Law provide conditional, distributed controls	Screen grant-backs, sourcing, territory and improvements under both regimes
Data and AI transfer	EU Data Act regulates defined access; EU AI Act allocates provider and downstream duties	Transfer law recognises data, software, models and algorithms; Data and AI Laws govern separate functions	Specify lawful access, permitted use, disclosure, records, incidents and role changes
Semiconductor capability	US CHIPS Act and EU Chips Act emphasise finance, facilities, pilot lines and skills	Sectoral incentives and strategic-technology rules coexist with treaty-constrained compulsory licensing	Combine voluntary licences with training, facilities and transparent access terms

Table 4 Functional international comparison and design implications

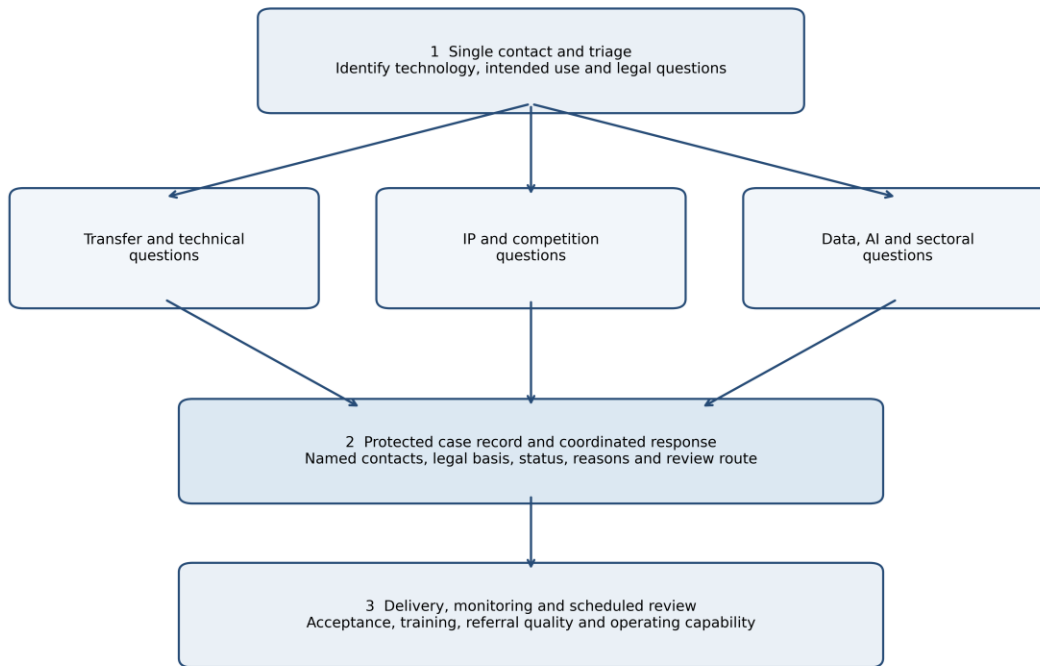


Fig. 4 Proposed coordination pilot. It is not an existing procedure or a single approval route

6 Discussion and testable implications

The study's broader claim is that regulatory resilience depends on usable connections among specialised functions. Multiple statutes are not inherently dysfunctional. Fragmentation becomes an empirical condition when participants cannot identify responsibilities, relevant decisions arrive too late for performance, information must be repeatedly reconstructed, or authorities apply incompatible classifications without a reviewable route. This definition permits comparison across jurisdictions without treating an emerging-economy label as an explanation.

The framework yields two propositions. First, if uncertainty about delivery is consequential, otherwise comparable transfers with explicit performance tests, correction procedures and payment milestones should exhibit fewer unresolved delivery disputes than transfers relying on generic descriptions. Second, if coordination is consequential, projects receiving earlier and reasoned referral among relevant authorities should experience fewer repeated compliance requests and late classification changes.

These propositions require data not available in the current corpus: contract files, acceptance records, administrative correspondence, judgments and interviews. Well-resourced parties may both draft better contracts and complete transfers more successfully. Complex technologies may generate more referrals and disputes regardless of administrative quality. Cases in which general law resolves a dispute despite sparse technology-specific drafting, or extensive contractual detail fails to prevent breakdown, are essential contrary cases.

The capability mechanism supplies an additional rival explanation. Finance, skills, facilities and organisational incentives may remain binding even when legal guidance is clear. Evaluation should therefore separate processing measures from operating outcomes. Referral time and repeated requests are coordination indicators;

acceptance, training completion and the recipient's capacity to operate the technology are performance indicators. Neither alone establishes economic or social value.

The Vietnamese case is analytically transferable where three conditions recur: performance depends on continuing cooperation, the transaction crosses regulatory mandates, and implementation depends on complementary public or organisational capacity. The particular clauses, agencies and statutory boundaries will differ. Transferability lies in the questions and mechanisms, not in copying Vietnam's legal architecture or assuming that all emerging economies share its constraints.

The research design also offers a methodological lesson. A small mixed-genre corpus can make agenda-setting transparent, but it cannot convert published argument into representative evidence. Sensitivity to genre is a substantive result because it shows which questions are carried by legal scholarship rather than public reporting. The correct response is to qualify the inference and connect it to doctrinal analysis, not to claim that model agreement validates a legal conclusion.

7 Conclusion

Technology-transfer systems must do more than define transferable objects and allocate rights. They must help a continuing relationship reach demonstrable operating capability while preserving the distinct mandates that govern intellectual property, competition, data, artificial intelligence and sectoral risk. Vietnam's recent reforms provide extensive distributed and conditional protection, but they leave important connective functions to contractual design and administrative practice.

The article's principal contribution is a dual-function account of the transfer contract. The contract is both a private performance arrangement and a compliance protocol. The two functions may rely on shared records and responsibilities, but their legal effects remain separate: private agreement cannot alter public powers or mandatory duties. A time-limited coordination pilot provides a way to test whether this connective role reduces duplication and late classification without creating another approval layer.

Three mechanisms organise the findings. Verifiable delivery links acceptance, correction, payment and warranty. Cross-regime coordination distinguishes transferability from ownership, lawful access and downstream compliance. Complementary capability explains why licences and definitions cannot substitute for skills, facilities, finance, extension institutions or valuation expertise. These mechanisms suggest different interventions and prevent legal-design problems from being conflated with implementation failure.

The limitations are substantial. Thirty-four publications cannot establish prevalence, causation or institutional performance. The contract-law grouping is small and genre-dependent. The study does not examine transaction files or evaluate the pilot. Its contribution is a bounded legal map, a mechanism-based framework and an empirical research programme. Subsequent work should test the propositions against contracts, acceptance records, administrative correspondence and judgments after the 2025–2026 reforms.

The legal map is current to 1 October 2026. Future use must check commencement provisions, implementing instruments and transaction-specific scope. That discipline is part of the article's substantive point: regulatory resilience depends not only on the existence of rules, but on accurate, timely and usable connections among them.

Data availability

An anonymized replication package is supplied as Supplementary Information. It contains the corpus inventory, supplementary model settings, scripts, document-topic assignments, K-scan outputs and robustness summaries used for the sensitivity audit and Fig. S1. Copyrighted full texts are not redistributed. The public

repository URL is provided to the editor in the title-page and cover-letter files and will be inserted in the manuscript after anonymous review.

Legal and policy instruments

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